# Chapter 13. Identifiable Gaussian Linear SEMs

Graphical models, Lecture 6 (equal error variances, conditional-variance ordering, ANM, the US algorithm)

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# 1 In General Gaussian SEMs, Directions Are Not Identified

$$\boxed{X \to Y \to Z} \qquad \boxed{X \leftarrow Y \to Z} \qquad \boxed{X \leftarrow Y \leftarrow Z}$$

The three structures can produce the same conditional independences and the same Gaussian distribution (Ch. 9).

| General Gaussian SEM: $\epsilon_j \sim N(0, \sigma_j^2)$ | Equal-variance Gaussian SEM: $\epsilon_j \sim N(0, \sigma^2)$ |
|---|---|
| if each variable may have its own error variance, several directions are compatible with the same distribution<br>result: CPDAG | if all structural errors are restricted to the same variance, the residual-variance structure differs by direction<br>result: DAG |

LiNGAM broke the tie with non-Gaussianity (Ch. 12). This chapter shows that, even for Gaussian data, an assumption on the *error variances* can identify the full DAG.

# 2 The Equal-Variance Gaussian SEM

**Definition: equal-variance Gaussian SEM**

Each variable is a linear combination of its parents plus a structural error:

$$X_j = \sum_{k \in \mathrm{pa}(j)} b_{jk} X_k + \epsilon_j, \qquad \epsilon_j \overset{\text{ind}}{\sim} N(0, \sigma^2),$$

$$X = BX + \epsilon, \qquad \Sigma = \sigma^2 (I - B)^{-1} (I - B)^{-\top}.$$

| Assumption | Meaning |
|---|---|
| linearity | effects of the parents are linear combinations |
| acyclicity | with a suitable ordering, $B$ is strictly lower triangular |
| independent errors | no hidden common cause among the analyzed variables |
| common error variance | all structural errors have the same $\sigma^2$ |
| nonzero edges | coefficients of the edges in the graph are not 0 |

Under these assumptions $B$ and the DAG are identified from the observational distribution.

# 3 Two Variables: The Equal-Variance Assumption Picks the Direction

**True direction** $X \to Y$:

$$X = \epsilon_X, \qquad Y = bX + \epsilon_Y, \qquad \epsilon_X, \epsilon_Y \overset{\text{ind}}{\sim} N(0, \sigma^2),$$

so the two structural errors have $\mathrm{Var}(\epsilon_X) = \mathrm{Var}(\epsilon_Y) = \sigma^2$.

**Reverse direction** $Y \to X$: rewrite the same distribution as $Y = \epsilon'_Y$, $X = cY + \eta$. Then

$$\mathrm{Var}(\epsilon'_Y) = (1 + b^2)\sigma^2, \qquad \mathrm{Var}(\eta) = \frac{\sigma^2}{1 + b^2}.$$

**Proof [Supplement].** $\mathrm{Var}Y = b^2\sigma^2 + \sigma^2$. The reverse regression has $c = \mathrm{Cov}(X,Y)/\mathrm{Var}Y = b/(1+b^2)$ and residual variance $\mathrm{Var}(X \mid Y) = \sigma^2 - \dfrac{b^2\sigma^4}{(1+b^2)\sigma^2} = \dfrac{\sigma^2}{1+b^2}$. ■

The two reverse error variances are equal only if $(1+b^2)^2 = 1$, i.e. $b = 0$. **Both directions explain the Gaussian distribution, but only the true direction satisfies the common error variance.**

# 4 Conditional Variances Reveal the Causal Order

**Motivation.** If all noise variances are equal, a source variable has the smallest variance, and after conditioning on it the next source has the smallest conditional variance. Repeating this recovers the causal order of a Gaussian SEM.

## 4.1 The key lemma

**Theorem: conditional-variance ordering**

In an equal-variance Gaussian SEM, let $S$ be an *ancestral* set (it contains all ancestors of each of its members, e.g. the first few variables of a causal order) and $j \notin S$. Then

$$\mathrm{Var}(X_j \mid X_S) = \sigma^2 \quad \text{if } \mathrm{pa}(j) \subseteq S, \qquad \mathrm{Var}(X_j \mid X_S) \geq \sigma^2(1 + b_{jk}^2) > \sigma^2 \quad \text{otherwise,}$$

for a suitable parent $k \notin S$ of $j$.

**Proof [Supplement].** Write $X = A\epsilon$ with $A = (I - B)^{-1}$; $a_{jl}$ is the total effect of $l$ on $j$ (sum over directed paths of products of coefficients), $a_{jj} = 1$, and $a_{jl} = 0$ unless $l$ is an ancestor of $j$ or $l = j$. Because $S$ is ancestral, every $X_s$, $s \in S$, is a function of $(\epsilon_l)_{l \in S}$ only, and conversely $(\epsilon_l)_{l \in S}$ is recovered from $X_S$ (invert the triangular system). Hence, with $X_j = \sum_{l \in S} a_{jl}\epsilon_l + \sum_{l \notin S} a_{jl}\epsilon_l$, the second sum is independent of $X_S$ and the first is determined by it:

$$\mathrm{Var}(X_j \mid X_S) = \sigma^2 \sum_{l \notin S} a_{jl}^2 = \sigma^2\Big(1 + \sum_{l \notin S,\, l \neq j} a_{jl}^2\Big).$$

If $\mathrm{pa}(j) \subseteq S$, every ancestor of $j$ is in $S$ (ancestral), so the sum is empty. Otherwise let $k$ be the parent of $j$ outside $S$ that is *latest* in a causal order. Any directed path from $k$ to $j$ other than the edge $k \to j$ ends with an edge $m \to j$ where $m$ is a parent of $j$ and a descendant of $k$; such an $m$ is not in $S$ ($S$ is ancestral and $k \notin S$) and is later than $k$, contradicting the choice of $k$. So $a_{jk} = b_{jk} \neq 0$, giving the bound. ■

So, given the already ordered variables, a *source* of the remaining graph has the smallest possible conditional variance $\sigma^2$, and every non-source has strictly more. This proves:

**Theorem: identifiability (Peters and Bühlmann, 2014)**

The DAG and $B$ of an equal-variance Gaussian SEM are identified from $\Sigma$. A causal order is obtained top-down: $\pi_1 = \arg\min_j \mathrm{Var}(X_j)$, and $\pi_m = \arg\min_{j \notin S} \mathrm{Var}(X_j \mid X_S)$ with $S = \{\pi_1, \ldots, \pi_{m-1}\}$.

### 4.2 A worked example

**Example: three variables**

Consider the Gaussian SEM

$$X_1 = \epsilon_1, \qquad X_2 = 0.8X_1 + \epsilon_2, \qquad X_3 = 0.7X_2 + \epsilon_3, \qquad \epsilon_1, \epsilon_2, \epsilon_3 \overset{\text{ind}}{\sim} N(0,1).$$

| Step | Variance comparison | Choice |
|---|---|---|
| 1 | $\mathrm{Var}(X_1) = 1$, $\mathrm{Var}(X_2) = 1.64$, $\mathrm{Var}(X_3) = 1.8036$ | $X_1$ |
| 2 | $\mathrm{Var}(X_2 \mid X_1) = 1$, $\mathrm{Var}(X_3 \mid X_1) = 1.49$ | $X_2$ |
| 3 | the last remaining variable | $X_3$ |

Here $\mathrm{Var}(X_2) = 0.64 + 1$, $\mathrm{Var}(X_3) = 0.49 \times 1.64 + 1$, and $\mathrm{Var}(X_3 \mid X_1) = 0.49\,\mathrm{Var}(X_2 \mid X_1) + 1 = 1.49$. Result: $X_1 < X_2 < X_3$.

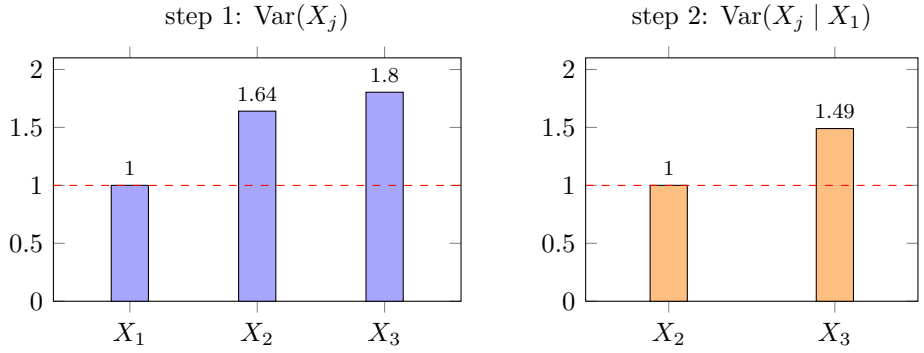



**Figure 1:** Top-down ordering in the equal-variance example ($\sigma^2 = 1$, red dashed line). At each step the variable whose (conditional) variance equals the common error variance — a source of the remaining graph — is selected: first $X_1$, then $X_2$.

### 4.3 After the order: select parents and edges

Let the estimated order be $X_1 < X_2 < X_3$. Each variable uses only earlier variables as candidate parents:

$$X_2 = \beta_{21}X_1 + \epsilon_2, \qquad X_3 = \beta_{31}X_1 + \beta_{32}X_2 + \epsilon_3.$$

Regression or sparse regression gives $\hat{\beta}_{21} \approx 0.8$, $\hat{\beta}_{31} \approx 0$, $\hat{\beta}_{32} \approx 0.7$, hence

$$X_1 \to X_2 \to X_3.$$

Conditional variances find the order of the variables; regression coefficients select the direct edges within that order.

## 5 Equal Variance Does Not Mean Equal Observed Variances

| Misconception | Correct meaning |
|---|---|
| all $X_j$ must have the same variance | variation transmitted from the parents makes observed variances differ (as in the example) |
| standardizing all variables gives equal variance | standardization rescales the structural errors too, so it does not guarantee the equal-variance assumption — it typically destroys it |
| similar fitted residual variances prove the assumption | only a diagnostic compatible with the chosen DAG; it does not prove the true direction |

When the equal-variance assumption is relatively natural: variables of the same kind and unit, whose measurement errors are also of similar size (e.g. expression levels measured on a common scale).

**Remark: scale dependence and varsortability**

The assumption is not scale-invariant: multiplying one variable by a constant changes its error variance. In many simulated benchmarks marginal variances simply increase along the causal order ("varsortability"), so methods exploiting variances can look excellent there; on real data with arbitrary units this signal may be absent or misleading. Check the variable units and measurement error, and the plausibility of the assumption, before trusting the order.

# 6 Additive Noise Models and the US Algorithm

## 6.1 The equal-variance Gaussian SEM is a special ANM

In an additive noise model each variable is

$$X_j = f_j(X_{\mathrm{pa}(j)}) + \epsilon_j, \qquad \epsilon_1, \dots, \epsilon_p \text{ independent.}$$

With $S$ the set of already selected variables, define the expected conditional variance

$$U_j(S) = \mathbb{E}\{\mathrm{Var}(X_j \mid X_S)\}.$$

If $S$ is ancestral and $j \notin S$, then $\epsilon_j \perp\!\!\!\perp X_S$ and

$$U_j(S) = \sigma_j^2 + \mathbb{E}[\mathrm{Var}\{f_j(X_{\mathrm{pa}(j)}) \mid X_S\}].$$

| Conditioning set $S$ | Conditional variance |
|---|---|
| all parents are in $S$ | $U_j(S) = \sigma_j^2$ |
| some parent not yet in $S$ | uncertainty created by the parents is added |

## 6.2 ANMs are identified by an ordering of conditional variances

Let the true causal order be $\pi = (\pi_1, \dots, \pi_p)$ and $S_{m-1} = \{\pi_1, \dots, \pi_{m-1}\}$. If at each step the true next variable has a smaller conditional variance than the other candidates,

$$U_{\pi_m}(S_{m-1}) < U_k(S_{m-1}), \qquad k \notin S_{m-1} \cup \{\pi_m\},$$

then the variables can be identified one by one.

| Equal-variance Gaussian SEM | General ANM |
|---|---|
| $\sigma_1^2 = \cdots = \sigma_p^2$<br>a variable whose parents are all included has no extra uncertainty, hence the smallest conditional variance (lemma above) | $\sigma_1^2, \dots, \sigma_p^2$ may differ<br>identified as long as the order of conditional variances, combining error variance and parent-induced uncertainty, is maintained |

The identifiability of ANMs needs neither linearity nor Gaussianity, but the *correct conditional-variance order*. (For general ANMs with nonlinear $f_j$ there are other identifiability results, based on residual independence as in LiNGAM; the conditional-variance route is the one used here.)

## 6.3 The US algorithm

**Algorithm: US — order by conditional variances, then parents**

**Step 1: estimate the causal order.** Let $S$ be the selected set (initially $\varnothing$).

1. Regress each $X_j$, $j \notin S$, on $X_S$ (flexibly, if $f_j$ may be nonlinear).
2. Compute the residual variance $\widehat{U}_j(S)$.
3. Select the variable with the smallest value, $\hat{\pi}_m = \arg\min_{j \notin S} \widehat{U}_j(S)$.

Add the selected variable to $S$ and repeat.
**Step 2: select the parents.** The candidate parents of each variable are the variables before it in the order. For each candidate, test conditional dependence given the other preceding variables:

$$X_{\pi_m} \not\perp\!\!\!\perp X_{\pi_j} \mid X_{\{\pi_1,\ldots,\pi_{m-1}\}\setminus\{\pi_j\}}.$$

If conditional dependence remains, add the edge $\pi_j \to \pi_m$.

US finds the order of directions with conditional variances and selects the actual edges with conditional independence. Within a correct order, the preceding variables other than the parents are non-descendants, so the Markov property guarantees that non-parents are conditionally independent given the rest (and faithfulness that parents are not).

# 7 Summary

| Model | Extra assumption | Clue for the direction |
|---|---|---|
| LiNGAM | linear relations with independent non-Gaussian errors | independence of the cause and the regression residual |
| equal-variance Gaussian SEM | common variance of independent Gaussian errors | order of conditional variances starting from the source |
| ANM and US | an additive noise model that keeps the conditional-variance order | conditional variances find the order; conditional independence selects the parents |

ANM and US allow nonlinear functions and different error variances, but still need acyclicity, no hidden confounding and causal minimality.

**Key points.**

- A general Gaussian linear SEM cannot distinguish Markov-equivalent directions from observational data.
- If all structural errors have the same variance, the full DAG is identified from the observational distribution.
- The source has the smallest variance; repeatedly comparing conditional variances yields the causal order.
- After the order, regression or sparse regression estimates the parents and edge coefficients.
- Check variable scales, measurement error, hidden confounding and violations of the equal-variance assumption.
- The complete DAG is obtained not from Gaussianity alone but from the additional assumption of a common structural error variance.